

NEO Mathematics Supporting Adult Guidance

Lesson 2 - Finding the Hypotenuse

Connection before curriculum. Always.

Purpose of the lesson

This lesson focuses on a narrow, secure skill: finding the hypotenuse when the two shorter side lengths are known. The mathematical priority is not speed. It is accurate orientation, correct side naming, and a visible solution route.

Adult role

- Help the learner enter the task calmly. The first move is always to find the right angle.
- Avoid beginning with the formula. Ask the learner to trace from the right angle to the opposite side.
- If the learner uses c automatically, pause and ask which capital vertex the hypotenuse is opposite. In this lesson the hypotenuse may be a , b or c .
- Encourage the Socratic Prompt before the Hint. Use the Hint as a re-entry tool, not as a correction.
- Protect dignity. A wrong dropdown choice is information about the learner's current model, not a failure.
- Use the Scratchpad, paper or oral explanation where appropriate.

Expected mathematical route

- 1 Identify the right-angle marker.
- 2 Name the side opposite the right angle as the hypotenuse.
- 3 Use the correct opposite-side letter: a opposite A , b opposite B , c opposite C .
- 4 Write the theorem with the hypotenuse square on the left.
- 5 Substitute the two shorter side lengths.
- 6 Square the shorter side lengths.
- 7 Add the squared values.
- 8 Take the square root to find the hypotenuse length.

Misconceptions to listen for

Learner action	Possible misconception	Helpful adult prompt
Chooses a side touching the right angle.	Hypotenuse is being read as a position or a visible slope, not as the side opposite the right angle.	Where is the right angle? Can you trace directly across to the opposite side?
Always puts c on the left.	The learner has memorised c as hypotenuse rather than using the opposite-side convention.	Which capital vertex is the right angle at? Which lower-case letter names the side opposite it?
Adds the two side lengths before squaring.	The learner is treating Pythagoras as a perimeter-like rule.	What does the theorem compare: lengths or square areas?
Stops at the squared total.	The learner has found the square of the hypotenuse, not the length.	What operation undoes squaring?

Learner action	Possible misconception	Helpful adult prompt
Accepts an answer shorter than one known side.	The learner is not sense-checking against the hypotenuse as longest side.	Should the hypotenuse be shorter or longer than the two sides meeting at the right angle?

Using Focus this step

The Focus this step button selects which step the Socratic Prompt and Hint buttons refer to. It is not a marking button. It helps the learner get targeted support without scrolling or interpreting a general hint.

Scratchpad standard

The full GreenPrint Scratchpad should be present: Pen -> Line -> Colour -> Width -> Eraser -> Undo -> Redo -> Sticky Note -> Background. Mathematical backgrounds should persist when learner marks are erased. The Line tool is especially useful for tracing the hypotenuse cleanly.

Support without over-supporting

- Give time for the learner to notice. Do not immediately name the hypotenuse for them.
- Use one-question prompts: 'Where is the right angle?' or 'What is opposite it?'
- Let the learner correct the dropdown choice after feedback.
- Celebrate a clear route as much as a correct answer.

Final evidence

A secure final task should show the right angle, the hypotenuse, the correct hypotenuse letter, a complete equation route, the square-root step, and a sense-check that the hypotenuse is longest.